



Efficient degradation of Levofloxacin with magnetically separable $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ Z-scheme heterojunction photocatalyst: Vis-NIR light response ability and mechanism insight

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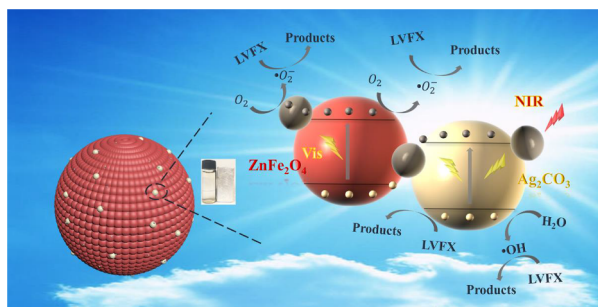
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HIGHLIGHTS

- A magnetically separable Z-scheme $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ photocatalyst was prepared.
- NCDs enhance charge separation and broaden solar light harvesting to NIR region.
- $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composites show excellent photoactivity for LVFX degradation.
- Possible photocatalytic degradation pathways of LVFX were proposed.

GRAPHICAL ABSTRACT



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ABSTRACT

A highly efficient and easily recoverable catalyst is always desirable for practical photocatalytic applications. In this study, a magnetically separable $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ Z-scheme heterojunction photocatalyst using nitrogen doped carbon quantum dots (NCDs) as solid-state electron mediators was fabricated by an easy-operated deposition-precipitation method. The prepared $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ photocatalyst exhibits a promoted photocatalytic activity for Levofloxacin (LVFX) removal compared to single ZnFe_2O_4 and Ag_2CO_3 , achieving 88.75% and 22.53% degradation efficiency under visible and near-infrared (NIR) light irradiation, respectively. Characterization results show that the synergistic effect of remarkable charge carrier separation efficiency and broadened solar light absorption is responsible for the enhanced photocatalytic performance. The well-matched band structures of the two semiconductors facilitate the charge transfer in Z-scheme transfer model, resulting in high charge separation rate. Simultaneously, due to the superb electron transfer character of NCDs, the introduction of NCDs can further accelerate the transfer of the photo-generated charges as solid-state electron mediators between ZnFe_2O_4 and Ag_2CO_3 . Furthermore, the unique up-converted photoluminescence properties of NCDs can broaden the optical absorption of the prepared $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ to NIR region, thus leading to an enhanced utilization of solar energy. This work provides a novel insight into the fabrication of novel all-solid-state Z-scheme photocatalyst based on nitrogen doped carbon quantum dots and opens new vista for the design of magnetically separable and highly efficient photocatalysts for environmental remediation.

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1. Introduction

Nowadays, antibiotics have been wildly employed to the treatment of bacterial infection due to their excellent antibacterial effect [1,2]. Levofloxacin (LVFX), possessing broad-spectrum antimicrobial properties and strong antibacterial, has been listed as one of the most commonly used antibiotics [3,4]. Frustratingly, the overuse of antibiotics results in their unexpected accumulation in aquatic environment, thus posing a serious threat on human health and aquatic ecosystems [5–7]. So far, many technologies have been put forward for the control of antibiotics in aquatic environment, such as photocatalysis [8], advanced oxidation process [9,10] and adsorption [11]. Among them, semiconductor-based photocatalysis has been considered to be one of the most appealing method due to its low-energy, environmental-friendly, and reusable properties. In addition, this technology has been put into widespread use in recent years for H_2 evolution, CO_2 reduction and organic contaminants degradation [12–15]. And the photocatalytic degradation for LVFX has been also investigated by many researchers. For example, Adhikari Sangeeta and co-authors achieved high-efficiency photocatalytic degradation of LVFX by using $MoS_2/Ag_2Mo_2O_7$ as photocatalyst, and the related photocatalytic mechanism for LVFX removal was also proposed [2,16,17]. However, the traditional photocatalysts are difficult to be separated from liquid phase, thus bringing secondary pollution to the aquatic environment, which is one of the primary problems in practical application process [13]. Now that the magnetic substrates exhibits remarkable recyclability under external magnetic field, the investigations on magnetically separable photocatalyst show great potential in the field of photocatalysis [18].

Spinel zinc ferrite ($ZnFe_2O_4$), a magnetic semiconductor with a relatively narrow band gap of 1.87 eV, has spawned ever growing interests in photocatalysis due to its excellent visible light response, extraordinary photochemical stability and easy recyclability [19,20]. Nonetheless, the ineffective charge separation and high recombination of photogenerated carriers severely restrict its photocatalytic performance in practical application [21]. In recent years, the design of hybrid photocatalysts with the construction of a heterojunction has been regarded as an effective strategy for preventing the recombination of photoinduced holes and electrons, which holds a strong redox ability at spatially separate sites [22,23]. And up to now, lots of $ZnFe_2O_4$ -based heterojunction composites with rapid charge separation efficiency and remarkable photocatalytic performance have been reported, such as $ZnFe_2O_4/Ag_2O$ [24], $Bi_2O_3CO_3/ZnFe_2O_4$ [25], $ZnFe_2O_4/ZnO$ [26] and so on. Inspired by above successful cases, combining $ZnFe_2O_4$ with other semiconductors to construct heterojunctions will be an effective method for boosting photocatalytic performance of $ZnFe_2O_4$.

Silver carbonate (Ag_2CO_3), with a suitable band gap of 2.3 eV, exhibits outstanding optical performance, which is usually used as a co-catalyst in photocatalyst [2]. Especially, due to its high valence band position (+2.92 eV vs. NHE), Ag_2CO_3 is always used to construct Z-scheme heterojunction to retain its strong oxidation potential and promote the transfer of photoinduced charges. For instance, Liu and coworkers synthesized a Z-scheme photocatalyst with extraordinary photocatalytic performance and good stability by modifying $g-C_3N_4$ with Ag_2CO_3 and graphene oxide [27]. Similarly, Wen et al. also found that with the decoration of Ag_2CO_3 and AgBr, the ternary $AgBr/CeO_2/Ag_2CO_3$ formed a unique double Z-scheme system, thereby holding superior photocatalytic activity [2]. Considering the advantage of Ag_2CO_3 and well-matched band positions between $ZnFe_2O_4$ and Ag_2CO_3 , it is rational to use Ag_2CO_3 as a co-catalyst to accelerate the separation rate of photoinduced charges of $ZnFe_2O_4$ with the construction of a Z-scheme heterojunction.

Over the past decade, the Z-scheme charge transfer mechanism has been put forward as a new charge transfer type and has been attracted numerous attentions. Compared to conventional type-II heterojunction photocatalyst, Z-scheme photocatalyst possesses stronger redox capacity due to its maximized utilization of more positive VB from one

catalyst and more negative CB from another one [28–32]. And the specific tandem configuration of the constructed Z-scheme photocatalyst provides an optimal energy to power each photocatalyst, which is beneficial for achieving efficient charge transportation and suppressing electron-hole recombination, thus enhancing the photoactivity of as-prepared composite. However, the competition between traditional charge transfer and Z-scheme charge transfer process always exists in some semiconductor heterojunctions, which will undoubtedly induce a negative impact on the photocatalytic activity [33]. Hence, introducing an electron transfer mediator between these two semiconductors, generally serving as a contact interface, is of vital significance for accelerating the transfer of photoinduced charge carries and promoting the construction of an efficient Z-scheme system [34].

Nitrogen doped carbon quantum dots (NCDs), a novel and eco-friendly zero-dimensional carbon nanomaterial with sizes below 10 nm, have drawn lots of attention because of their excellent electric and optical properties [35]. The conjugated π structure of NCDs not only brings them superb electron reservoir and transfer properties but also enhances the contact between NCDs and other semiconductors to establish stable composites [36]. What's more, NCDs with unique up-converted photoluminescence properties can act as photosensitizers for catching both visible and NIR light [37]. When they couple with other semiconductors, the composites can turn a NIR light into a visible light. These characteristics make NCDs an ideal candidate for modifying the semiconductors as an electron transfer mediator to improve their photocatalytic activities and the utilization of solar light. Therefore, it is anticipated that introduction of NCDs into binary $ZnFe_2O_4/Ag_2CO_3$ system can further promote their light response range and separation rate of photoinduced charges.

Herein, a magnetically separable $ZnFe_2O_4/NCDs/Ag_2CO_3$ all-solid-state Z-scheme photocatalytic system was successfully constructed by a mild deposition-precipitation process. In this hybrid structure, $ZnFe_2O_4$ and Ag_2CO_3 play the roles of separating photoinduced electron-hole pairs through the formed Z-scheme heterojunction, while NCDs act as the transport bridge for photogenerated carriers and photosensitizers for light absorption. The phase, chemical composition, morphologies, photoelectric and optical properties of the prepared catalysts were investigated in detail. The photocatalytic activity of the composite under both visible and NIR light were evaluated by Levofloxacin (LVFX) degradation and the main degradation intermediates were further analyzed by high performance liquid chromatography-mass spectra (HPLC-MS). Additionally, the cycle experiment and VSM were used to research the stability and magnetic behavior of the obtained $ZnFe_2O_4/NCDs/Ag_2CO_3$ photocatalyst. The radical trapping experiments and electron spin resonance (ESR) technology were carried out to detect the active species generated during reaction. In light of above experiment analysis, a plausible degradation mechanism of LVFX over $ZnFe_2O_4/NCDs/Ag_2CO_3$ was proposed.

2. Experimental

2.1. Preparation of the photocatalysts

2.1.1. Preparation of NCDs

All the reagents with analytical grade are used directly in this experiment. The NCDs aqueous was obtained according to the previous reported method [38]. Firstly, 1.05 g citric acid and 0.335 mL ethylenediamine were added into 10 mL deionized water. Then, the mixture was placed into a 50 mL Teflon-lined stainless autoclave and sealed at 200 °C for 5 h. Finally, above acquired liquid was centrifuged for removing large particles and then followed by freeze drying. The acquired NCDs solid were dissolved into deionized water to get uniform NCDs aqueous (1 mg/mL). The resultant dark yellow solution was water soluble NCDs.

2.1.2. Preparation of magnetic ZnFe_2O_4

Magnetic ZnFe_2O_4 was fabricated through a hydrothermal and calcined method [24]. In a typical process, 6 mmol $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 3 mmol $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ were dissolved in 35 mL ethylene glycol. Then 10 mL ethylene glycol solution containing 9 mmol sodium acetate was dropped into the above mixture and kept stirring for 30 min at ambient temperature. Subsequently, the suspension was heated at 180 °C in a 100 mL Teflon-lined stainless autoclave for 18 h. The resulting black precursor then turned to orange by calcining at 500 °C.

2.1.3. Preparation of $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$

$\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ composites were obtained via a facile deposition-precipitation process. Shortly, 0.20 g ZnFe_2O_4 was dispersed into 60 mL deionized water with the assistance of ultrasound. Subsequently, a certain amount of 0.10 mol/L AgNO_3 aqueous was added into aforementioned mixture and kept stirring for 30 min. Then, excess 0.10 mol/L NaHCO_3 solution was dripped into the mixed under stirring in darkness for 2 h. The resulting precipitate was dried at 60 °C after rinsed with deionized water thoroughly. The obtained samples were denoted as $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ -17%, $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ -21%, $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ -25% (or $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$) and $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ -29% when the mass of Ag_2CO_3 accounts for 17%, 21%, 25% and 29% among entire composite, respectively.

2.1.4. Preparation of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$

In a typical synthesis process, a certain amount of NCDs (1 mg/mL) were dropped into 5.0 mL AgNO_3 solution (0.10 mol/L), after ultrasonication for 20 min, the homogeneous liquid was added to 60 mL deionized water with 0.20 g ZnFe_2O_4 dissolved. Followed by 30 min stirring, excess 0.10 mol/L NaHCO_3 solution was dripped into the mixed. After stirring in dark condition for 2 h, the precipitate was dried at 60 °C after rinsed with deionized water thoroughly. The as-prepared samples were denoted as $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ -1/ Ag_2CO_3 -25%, $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ -3/ Ag_2CO_3 -25%, $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ -5/ Ag_2CO_3 -25% (or $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$) and $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ -7/ Ag_2CO_3 -25% for the NCDs feeding amount of 1 mL, 3 mL, 5 mL and 7 mL, respectively.

In comparison, pure Ag_2CO_3 was synthesized in the absence of ZnFe_2O_4 and NCDs. The $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ -5 (or $\text{ZnFe}_2\text{O}_4/\text{NCDs}$) composite was synthesized by mixing 5 mL NCDs (1 mg/mL) with 0.20 g ZnFe_2O_4 in 60 mL deionized water. The $\text{Ag}_2\text{CO}_3/\text{NCDs}$ -5 (or $\text{Ag}_2\text{CO}_3/\text{NCDs}$) composite was prepared by the same method as the $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ in the absence of ZnFe_2O_4 , and the amount of NCDs were 5 mL.

2.2. Characterization

X-ray powder diffraction (XRD) of the prepared catalysts were performed on a Bruker D8 ADVANCE with the irradiation of $\text{Cu K}\alpha$. The morphology and crystal planes of the samples were identified by transmission electron microscopy (TEM) and high-resolution TEM (HRTEM, JEM-2010). The distributions of elements in photocatalyst were observed by energy-dispersive spectrometry (EDS). Chemical composition was investigated through the X-ray photoelectron microscopy (XPS) on Thermo ESCALAB 250XI spectrometer with $\text{Al K}\alpha$ source. A Shimadzu UV-2450 spectrometer was applied to record the UV-vis diffuse reflectance spectra (DRS). The Brunauer-Emmett-Teller (BET) specific surface area was investigated on TRISTAR-3000 analyzer. The mappings of 3D EEMs were collected from a Fluoromax-4 spectrofluorometer. The magnetic properties were carried out on a MPMS (SQUID) XL-7 vibrating sample magnetometer (VSM). The electron spin resonance (ESR) signals of superoxide and hydroxyl radicals were investigated on a Bruker model ESR E500 spectrometer.

2.3. Photocatalytic experiments

The photocatalytic performances of the synthesized catalysts were

assessed by the degradation of LVFX under both visible and NIR light irradiation. A 300 W Xe lamp (Zhong jiao jin yuan, CEL-HXF300) equipped with 420 nm and 760 nm cut filters was used as light source. Typically, 30 mg photocatalyst was put into LVFX (50 mL, 10 mg/L) solution. Before irradiation, the suspension was kept stirring for 30 min in darkness to achieve the equilibrium between adsorption and desorption. At a certain interval of irradiation, 3 mL suspension was collected after centrifugation remove the residual particles. The relevant concentration of residual LVFX was measured by an UV-vis spectrophotometer at its maximum absorption peak (288 nm). The UV-vis graph of different concentrations of LVFX is provided in [Supporting Information \(Fig. S1\)](#). The photodegradation intermediates of LVFX were investigated by LC-MS. Related details can be found in [Supporting Information](#).

2.4. Electrochemical measurements

The photoelectrochemical characteristics of the as-prepared materials were investigated in a conventional three-electrode model on an electrochemical workstation (CHI760D, Shang Hai, China). Pt wire and Ag/AgCl electrodes were acted as counter and reference electrodes, respectively, while the FTO with photocatalysts coated on as the working electrode. As for the preparation the working electrodes, 10 mg of the as-prepared catalyst was put into 0.1 mL of 25% PVA solution, the suspension after sonication was put onto FTO slice containing active area of 1 cm^2 and dried at 80 °C. During the photoelectrochemical test process, 0.2 mol/L sodium sulfate electrolyte solution was used as an electrolyte and a 300 W Xe lamp (Zhong jiao jin yuan, CEL-HXF300) equipped with 420 nm and 760 nm cut filters offered the light source. The photocurrent response was measured under an open-circuit potential. To further study the photoelectrochemical characteristics of the prepared samples, electrochemical impedance spectroscopy (EIS) was investigated on an electrochemical workstation (CHI760D, Shang Hai, China) equipped with a frequency analyzer module in the frequency range of 10^5 to 10^{-2} Hz. Finally, Mott-Schottky was measured with a frequency of 1 kHz.

3. Results and discussion

3.1. Physicochemical characterization

The phase purities and crystal structures of Ag_2CO_3 , ZnFe_2O_4 , $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composites were investigated by the X-ray diffraction (XRD) technique. Just as shown in [Fig. 1](#), the pure Ag_2CO_3 peaks appear at $2\theta = 18.61^\circ$, 20.60° , 32.63° , 33.70° , 37.82° and 39.62° , which can be indexed to the (0 2 0), (1 1 0), (-1 0 1), (-1 3 0), (2 0 0), and (0 3 1) planes, respectively (JCPDS No. 26-0339) [39]. Meanwhile, the six diffraction peaks of single ZnFe_2O_4 at $2\theta = 30.1^\circ$, 35.4° , 43.1° , 53.4° , 56.9° and 62.5° (JCPDS standard card 22-1012) correspond to the indexes (2 2 0), (3 1 1), (4 0 0), (4 2 2), (5 1 1) and (4 4 0), respectively [40]. And the broad peaks reveal its small crystal size. Apparently, all the major peaks of ZnFe_2O_4 and Ag_2CO_3 are preserved in the XRD pattern of $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$, demonstrating that the combination with Ag_2CO_3 and NCDs do not affect the crystal structure of ZnFe_2O_4 . However, there are no obvious diffraction patterns of NCDs, which may be ascribed to its low concentration and weak diffraction signal [35]. Simultaneously, no additional crystal phases are observed in all patterns, suggesting the high purity of the as-prepared catalysts.

The morphologies of the as-prepared composites were identified by TEM and HRTEM. As shown in [Fig. 2a](#), the pristine ZnFe_2O_4 exhibits a uniform spherical structure with diameters ranging from 100 nm to 150 nm, which will provide large surface area and considerable active sites for the photocatalytic degradation. As displayed in [Fig. 2c](#) and [d](#), Ag_2CO_3 nanoparticles and the small-sized NCDs are well anchored on the surface of ZnFe_2O_4 , forming a close contacted ternary composites.

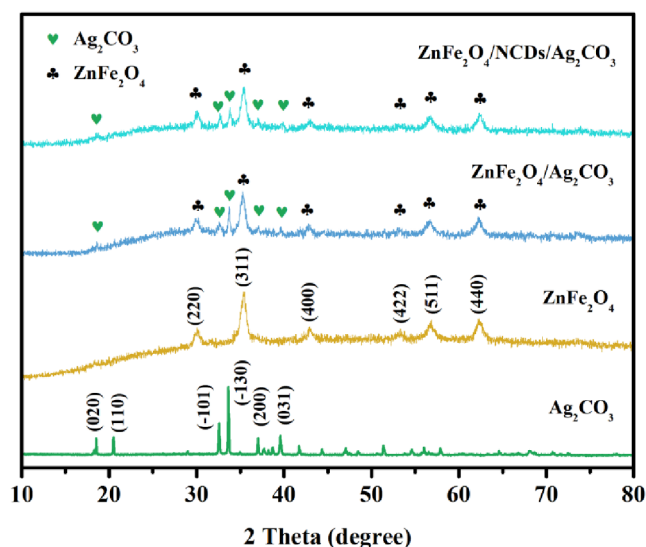


Fig. 1. XRD patterns of prepared Ag₂CO₃, ZnFe₂O₄, ZnFe₂O₄/Ag₂CO₃ and ZnFe₂O₄/NCDs/Ag₂CO₃ composites.

After the introduction of Ag₂CO₃ and NCDs, there are no obvious morphology change of ZnFe₂O₄, indicating its good structural stability. From the HRTEM images in Fig. 2d, we can find three different lattice spaces of 0.253, 0.209 and 0.33 nm, which can be assigned to (3 1 1) plane of ZnFe₂O₄, (1 3 1) plane of Ag₂CO₃ and (0 0 2) plane of NCDs, respectively. Additionally, SEM image (Fig. 2e) and corresponding EDS mapping (Fig. 2f~k) were performed to further investigate the appearance and element distribution of the ZnFe₂O₄/NCDs/Ag₂CO₃ nanocomposite. We can see from Fig. 2e that there are numerous well-dispersed ZnFe₂O₄ nanospheres decorated with little Ag₂CO₃ nanoparticles, which is completely in conformity with the results of TEM images. And EDS image of ZnFe₂O₄/NCDs/Ag₂CO₃ composite shows that six elements of Zn, Ag, O, Fe, N and C are observed in the sample, which is accord with the element composition of ZnFe₂O₄/NCDs/Ag₂CO₃ nanocomposite. Consequently, results above not only demonstrate the successful synthesis of the ZnFe₂O₄/NCDs/Ag₂CO₃ composite but also prove the close contact among the three phases, which contribute to the separation of the photoinduced charges and boost the ultimate photocatalytic activity.

The specific surface area and pore characteristic of the as-prepared samples were investigated by nitrogen adsorption and desorption isotherm measurements. As displayed in Fig. 3 and Table S1, the BET surface area (S_{BET}) of the ZnFe₂O₄/NCDs/Ag₂CO₃ nanocomposite ($29.5317 \text{ m}^2 \text{ g}^{-1}$) shows a slight decrease compared to that of pure

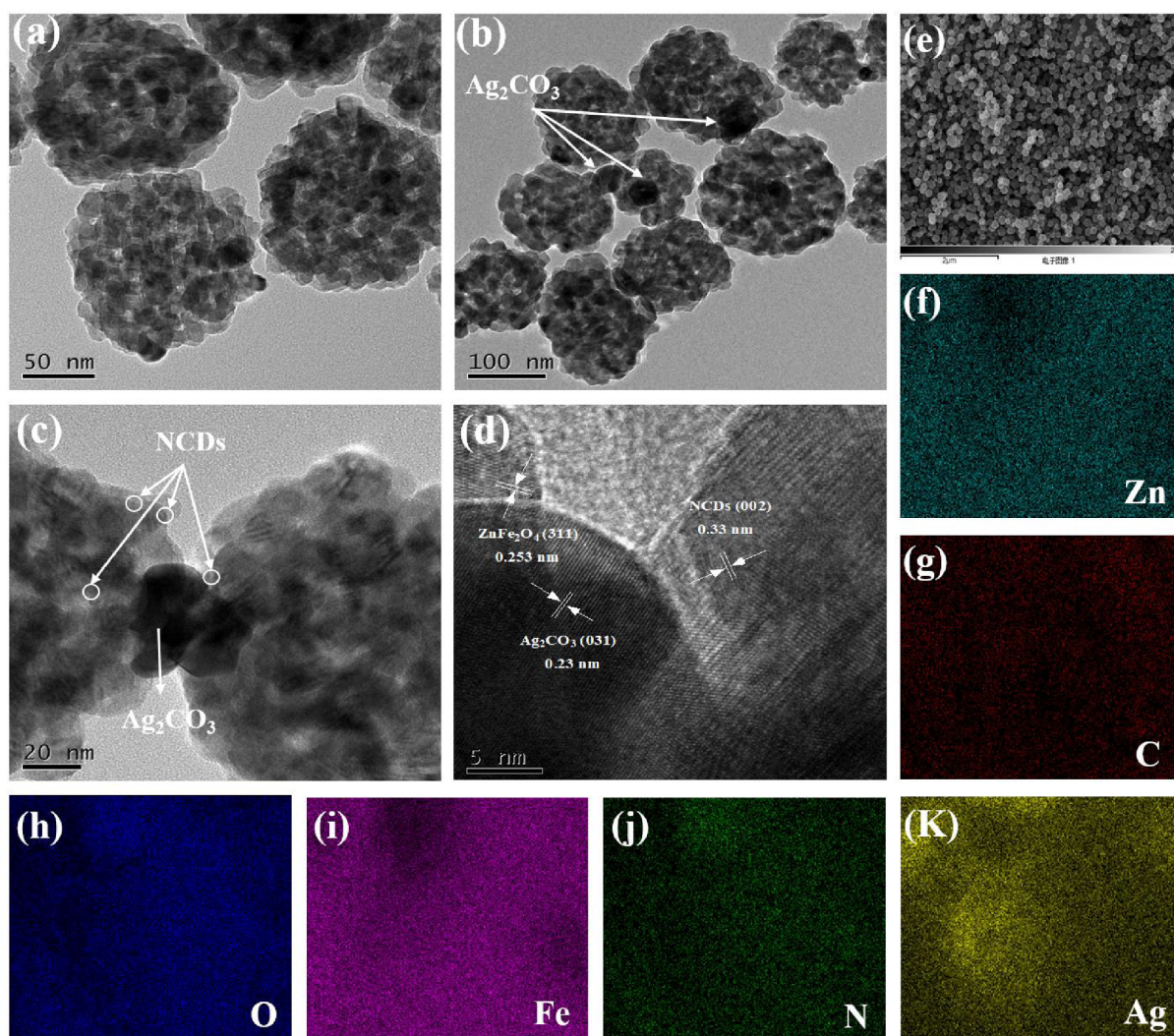


Fig. 2. TEM images of (a) ZnFe₂O₄, (b) ZnFe₂O₄/Ag₂CO₃ composite, (c) ZnFe₂O₄/NCDs/Ag₂CO₃ composite, (d) HRTEM and (e) SEM image of ZnFe₂O₄/NCDs/Ag₂CO₃ composite, (f–k) EDS elements mapping images of ZnFe₂O₄/NCDs/Ag₂CO₃ composite.

ZnFe_2O_4 ($33.3262 \text{ m}^2 \text{ g}^{-1}$). This may be attributed to the fact that the Ag_2CO_3 nanoparticles with a relatively small S_{BET} of $3.27 \text{ m}^2 \text{ g}^{-1}$ are dispersed on the surface of large sphere-like ZnFe_2O_4 substrate. Hence, the surface pore of ZnFe_2O_4 might be impeded by Ag_2CO_3 nanoparticles, resulting in a lower S_{BET} of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$. From the results of BET analysis, it can be inferred that the surface area is not the major factor determining the photocatalytic activity in this system.

To further investigate the surface chemical states and compositions of the as-prepared $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$, XPS analysis was carried out. It is clear in Fig. 4, the survey XPS spectrum shows that the main elements of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite are Zn, Fe, Ag, O, C and N, which is consistent with the results of EDS. Fig. 4b exhibits the high-resolution XPS spectra of Fe 2p with two main peaks at 711.3 eV and 724.2 eV, corresponding to Fe 2p_{3/2} and Fe 2p_{1/2}, respectively, which reveals that Fe element presents as Fe^{3+} state [41,42]. As shown in Fig. 4c, the peaks centered on 1020.9 eV and 1044.1 eV separately belong to the binding energy of Zn 2p_{3/2} and Zn 2p_{1/2}, suggesting the appearance of Zn^{2+} [43]. As for Ag 3d (Fig. 4d), there are two obvious peaks situating at binding energies of 367.8 eV and 374.1 eV, which can be ascribed to Ag 3d_{5/2} and Ag 3d_{3/2}. This result demonstrates Ag element presents in $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ nanocomposite in the form of Ag^+ [44]. The high-resolution C 1s XPS spectrum is displayed in Fig. 3e and two evident peaks are observed at peaks of 284.8 eV and 288.3 eV. The peak at 284.8 eV can be identified as C–C bond in NCDs [35]. Another peak centered on 288.3 eV belongs to C=O band in Ag_2CO_3 [45]. The O 1s peak (Fig. 4f) can be divided into three peaks. Two main peaks located at 529.7 eV and 530.4 eV correspond to oxygen species of the lattice oxygen of ZnFe_2O_4 and Ag_2CO_3 , respectively [43,46]. Another small one with the binding energy of 531.6 eV may derive from the surface hydroxyl group or the adsorbed H_2O on the surface of the catalysts [47]. And the N 1s peak situated at 401.1 eV comes from the N–H in the NCDs [48]. The XPS analysis further confirms the successful preparation of ZnFe_2O_4 , Ag_2CO_3 and NCDs.

For exploring the optical characteristics of the as-prepared composites, UV–vis diffuse reflectance spectra were conducted. As depicted in Fig. 5a, the pristine ZnFe_2O_4 nanospheres present a strong absorption edge at 710 nm, originating from the photo-inspired charge transfer from its valence band to the conduction band [49]. Compared to ZnFe_2O_4 , single Ag_2CO_3 exhibits a relatively low visible-light absorption in visible-light region, with a band gap edge around 530 nm [46]. The composite of $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ shows a mixed optical absorbance with the absorption edge located at 700 nm, indicating a strong interaction between ZnFe_2O_4 and Ag_2CO_3 . After the introduction of NCDs, an obvious red shift edge is observed, suggesting enhanced optical absorption of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite. The phenomenon is attributed to the uniform distribution of NCDs on the surface of ZnFe_2O_4 and Ag_2CO_3 , which strengthens their visible light harvesting capacity. It's known that NCDs show excellent upconversion photoluminescence properties according to previous reports [50]. Therefore, when NCDs couple with ZnFe_2O_4 and Ag_2CO_3 , the composites can transform the near infrared wavelength light into a shorter visible light, thereby promoting the light absorbing ability of the composites.

Furthermore, band gaps of pristine ZnFe_2O_4 and Ag_2CO_3 were calculated by the Kubelka-Munk function [51]:

$$\alpha h\nu = A(h\nu - E_g)^{n/2}$$

In this formula, α , h , ν , E_g , and A denotes the absorption coefficient, Planck constant, light frequency, band gap energy, and proportionality constant, respectively. In addition, n is related to the inner characteristic of semiconductor and it is 1 for pristine ZnFe_2O_4 and Ag_2CO_3 because they are both direct semiconductors. According to the $(\alpha h\nu)^2$ against $(h\nu)$ curve (Fig. 5b), band gap energies of ZnFe_2O_4 and Ag_2CO_3 are 1.87 and 2.57 eV, respectively.

3.2. Photocatalytic performance

3.2.1. Photocatalytic performance of different photocatalysts

LVFX was utilized as target pollutants to assess the photocatalytic activities of the prepared samples under both visible and NIR light irradiation. Prior to photocatalytic reaction, the adsorption–desorption equilibrium is achieved by continuously stirring for 30 min in the dark. As displayed in Fig. 6a, the blank test result suggests the self-degradation of LVFX is almost negligible. It is noticed that the single photocatalyst shows limited activities, which is because of undesirable charge carrier recombination. For pure ZnFe_2O_4 , the degradation efficiency for LVFX is only 11.82%, and it is 66.52% for pristine Ag_2CO_3 under the same condition. However, the binary and ternary composites exhibit substantially enhanced photocatalytic performance compared to pristine photocatalyst. We carried out a series of experiments to investigate the optimal ratio of ZnFe_2O_4 , NCDs and Ag_2CO_3 . The relative description and data are presented in the Supporting Information (Fig. S2). Among all samples, the ternary sample, $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$, achieves the highest degradation efficiency and approximately 88.75% of LVFX can be degraded within 90 min. The enhancement for degradation ability of the ternary sample can be put down to the reasons: (i) the fabricated Z-scheme heterojunction boosts the diffusion of photoinduced charges as well as interfacial charge transfer process; (ii) the introduction of NCDs not only expedites the separation of photo-induced charge carrier as a transport medium but also brings wider light absorption.

To fully explain the degradation process, the Langmuir-Hinshelwood model was applied and the reaction rate constants of the catalysts follow the sequence as $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3 > \text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3 > \text{Ag}_2\text{CO}_3/\text{NCDs} > \text{Ag}_2\text{CO}_3 > \text{ZnFe}_2\text{O}_4/\text{NCDs} > \text{ZnFe}_2\text{O}_4$, which well matches with the degradation efficiency. Obviously, $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composites achieve the highest reaction constant, 0.02081 min^{-1} , which is 1.87, 1.71, 18.58, 12.17 and 1.40 folds as high as that of Ag_2CO_3 , $\text{Ag}_2\text{CO}_3/\text{NCDs}$, ZnFe_2O_4 , $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ and $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ respectively. This result further demonstrates that ZnFe_2O_4 couples with Ag_2CO_3 and NCDs can boost the separation rate of photoinduced holes and electrons, and thus promote the photocatalytic performance.

In addition, the photocatalytic performance of the synthesized ZnFe_2O_4 , $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ have been

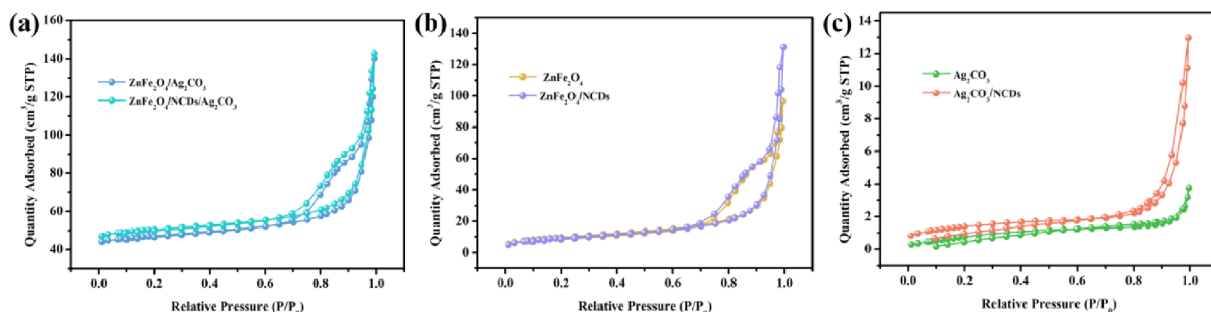


Fig. 3. N_2 adsorption–desorption isotherm of prepared samples.

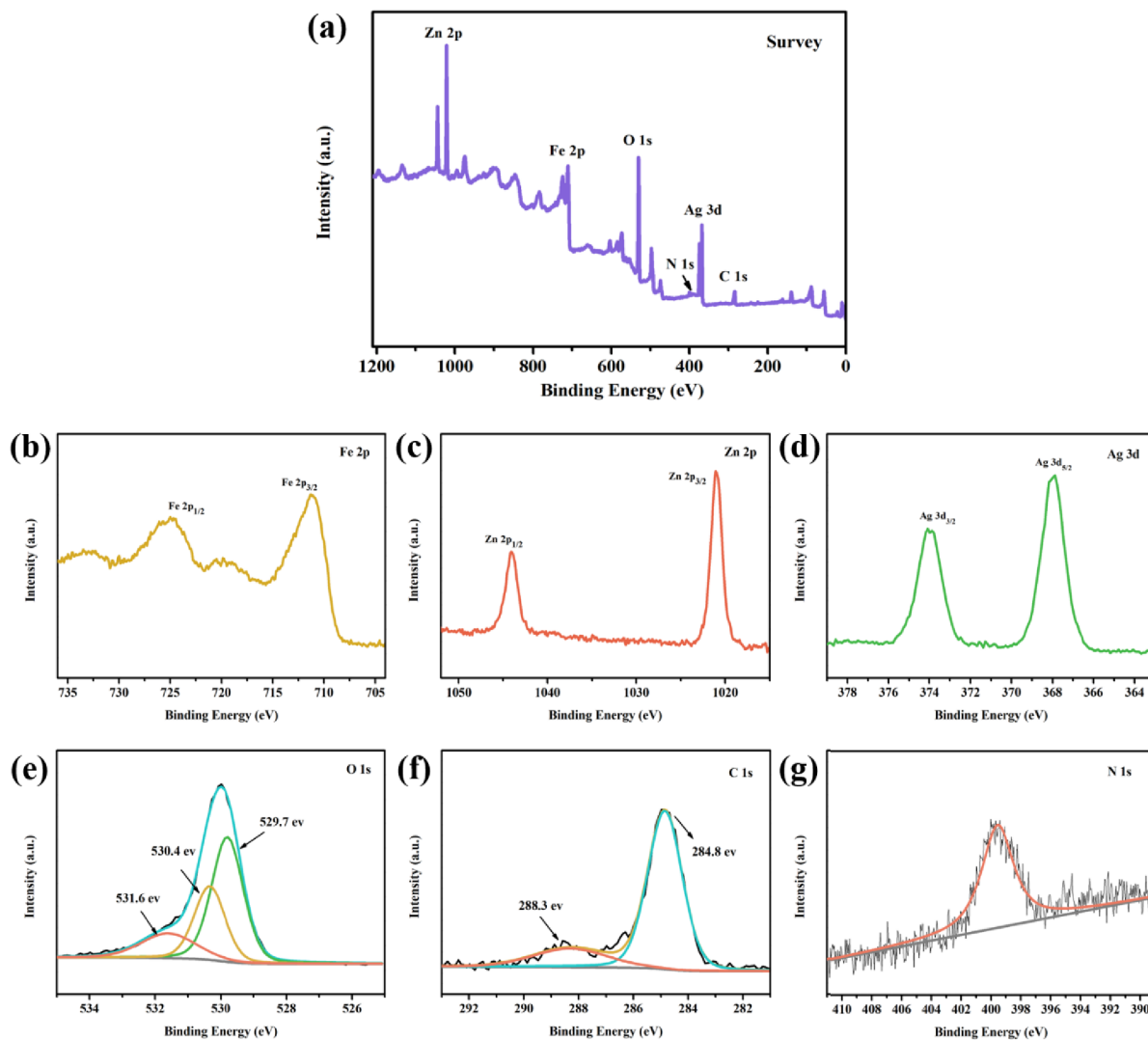


Fig. 4. XPS spectra of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite: (a) the survey spectrum, (b) Fe 2p, (c) Zn 2p, (d) Ag 3d, (e) O 1s, (f) C 1s and (g) N 1s.

further studied under the NIR light irradiation. As Fig. 7a shown, the photoactivity of pristine ZnFe_2O_4 can be neglected under NIR light irradiation because of its low photon energy. Nevertheless, the photocatalytic performance of $\text{ZnFe}_2\text{O}_4/\text{NCDs}$ is promoted compared to single ZnFe_2O_4 due to the upconverted photoluminescent properties of NCDs, which can convert NIR light to visible light so that the photocatalytic reaction system can produce efficient active species. Furthermore, the as-prepared ternary $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite

shows the highest photocatalytic performance among all composites, presenting a 22.53% degradation efficiency of LVFX under NIR light irradiation within 180 min, which may be ascribed to the enhanced charges separation rate and the broadened solar light absorption, resulting from the synergistic effect of formed Z-scheme heterojunction as well as upconverted photoluminescent character of NCDs. Meanwhile, from the Fig. 7b, we can see that the rate constant of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ is $0.000561 \text{ min}^{-1}$, which is 11.6 and 3.92 times higher than

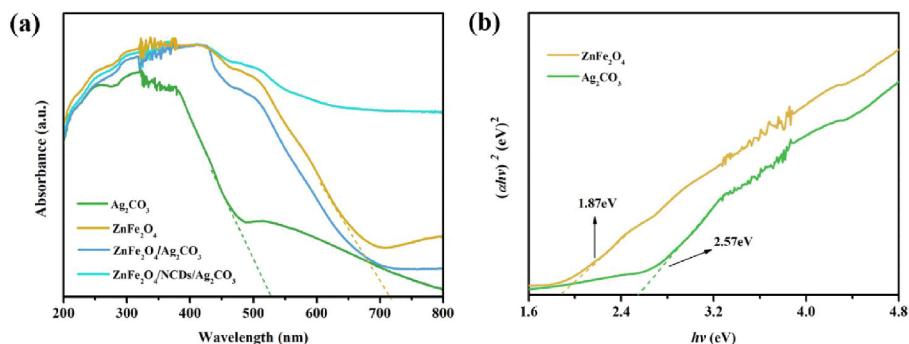


Fig. 5. (a) DRS spectra of as-prepared Ag_2CO_3 , ZnFe_2O_4 , $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite; (b) Plots of $(\alpha h\nu)^2$ vs. $(h\nu)$ of pure Ag_2CO_3 and ZnFe_2O_4 .

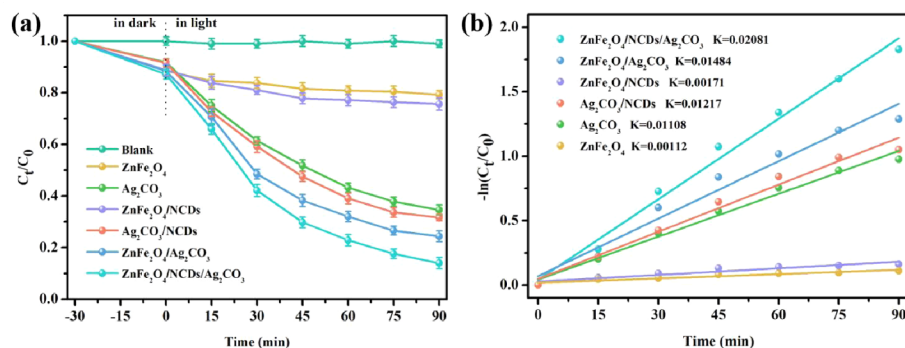


Fig. 6. (a) Photoactivity of as-synthesized photocatalysts for LVFX removal under visible light and (b) first-order kinetics plot of as-prepared samples.

that of pristine $ZnFe_2O_4$ and $ZnFe_2O_4/NCDs$. The results above reveal that the ternary $ZnFe_2O_4/NCDs/Ag_2CO_3$ composite shows high photocatalytic activity under both visible light and NIR light irradiation.

3.2.2. Effect of initial LVFX concentrations

In natural environment, the concentration of LVFX varies extensively. Therefore, the investigation for the influence of different initial LVFX concentration on photocatalyst's performance is necessary. From Fig. 8a, it is known that the initial concentration of LVFX has a great impact on photocatalytic degradation reaction and the degradation efficiency of LVFX is decreased from 88.75% to 45.65% as the concentration increases from 10 mg/L to 40 mg/L. Actually, intermediates would be produced in high concentration LVFX solution during photodegradation process, which are the strong competitors for LVFX molecules to get the limited active adsorption and catalytic sites. Moreover, higher initial pollutant concentrations could restrain the penetration of light into aqueous solution, resulting in the reduction of transfer path of photons and available photons [52]. Therefore, the concentration of LVFX is selected as 10 mg/L during the whole experiment for eliminating the negative effect and achieving optimal photocatalytic activity.

3.2.3. Effect of catalyst dosage

The optimal dosage of catalysts is of great importance for reducing industrial production cost and achieving high photocatalytic efficiency in practical application. Thus, photocatalytic activities of LVFX with different catalyst dosage in the range of 0.20–1.40 g/L were performed to find out the optimal dosage. Fig. 8b shows that the degradation efficiency of LVFX is significantly improved from 64.52% to 88.75% with the amount of $ZnFe_2O_4/NCDs/Ag_2CO_3$ photocatalyst increasing from 0.20 g/L to 0.60 g/L. However, as the photocatalyst dosage further increase, the degradation efficiency of LVFX decrease gradually. The reasons of the observed phenomenon may put down to the following two. On the one hand, the number of photons is a constant under the irradiation of visible light in a certain time and the low dosage of

photocatalyst results in low LVFX degradation efficiency due to the insufficient utilization of photos [53]. Therefore, there should be an optimal dosage of photocatalyst to make full use of the photo energy. On the other hand, the turbidity of solution can be increased by the increasing dosage of catalyst, which reduces light penetration and thus inhibits the photocatalytic ability. Hence, the catalyst dosage of 0.60 g/L is chosen as optimal dosage for the whole experiment.

3.2.4. Effect of supporting electrolyte

In actual water bodies, there are many inorganic ions that can interfere with the removal of target pollutants. Research on the influence of the co-existed anions on LVFX removal in $ZnFe_2O_4/NCDs/Ag_2CO_3$ system can provide theoretical basis for practical applications. Hence, typical cations such as Na^+ , Ca^{2+} , Mg^{2+} and anions such as Cl^- , SO_4^{2-} , NO_3^- were added into photodegradation system at a concentration of 5 mM to evaluate their effects on the LVFX removal. As shown in Fig. 8c, the addition of Na^+ has a slight effect on photodegradation process, which can be neglected, while both Ca^{2+} and Mg^{2+} show significant inhibition for the degradation of LVFX and the addition of Mg^{2+} presents a more obvious negative effect on the degradation of LVFX than Ca^{2+} . The decrease in photocatalytic performance could be explained by the fact that both Ca^{2+} and Mg^{2+} are able to form metal complexes with LVFX, making LVFX difficult to degrade [37]. In addition, the more obvious inhabitation of Mg^{2+} is attributed to its stronger coordination ability. As depicted in Fig. 8d, the addition of SO_4^{2-} and NO_3^- cause negligible effect on the degradation process. The presence of Cl^- poses slight decrease in photodegradation process, which possibly puts down to the inevitable competitive adsorption between Cl^- and LVFX at the active sites of catalyst [12].

3.2.5. Effect of different contaminants

To further investigate the photocatalytic activity of the $ZnFe_2O_4/NCDs/Ag_2CO_3$ catalyst towards different contaminants, photodegradation experiments for Oxytetracycline hydrochloride (OTC), Ciprofloxacin (CIP) and Enrofloxacin (EFA) were carried out under

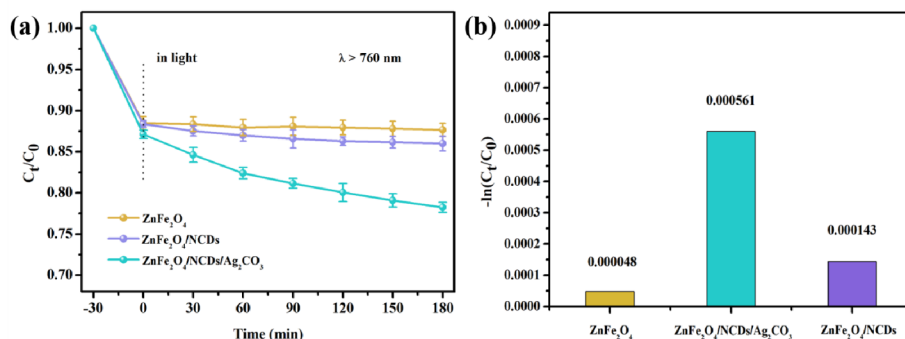


Fig. 7. (a) Photoactivity of as-synthesized photocatalysts for LVFX removal under NIR light and (b) kinetic constants.

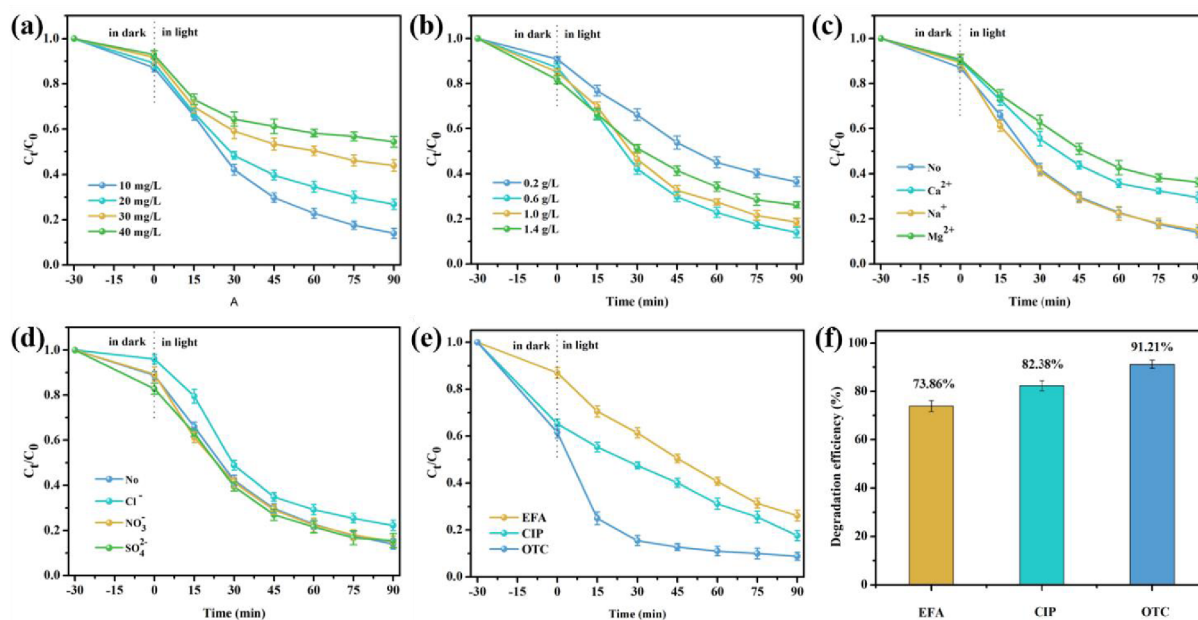


Fig. 8. Influence of (a) initial concentration of LVFX (b) catalysis dosage, (c) and (d) supporting electrolytes on degradation of LVFX over $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite. The photocatalytic activity (e) and degradation efficiency (f) of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ towards other contaminants.

visible light. With the irradiation time of 90 min, the degradation efficiency of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ towards OTC (20 mg/L), CIP (10 mg/L) and EFA (10 mg/L) are 91.21%, 82.38% and 73.86%, respectively (Fig. 8f). The results reveal that the as-prepared photocatalyst exhibits remarkable photocatalytic performance for the removal of antibiotics.

3.3. Possible degradation pathway of LVFX

LC-MS technique was used to further detect intermediates produced in photodegradation reactions, which contributes to an in-depth understanding of the mineralization pathways of LVFX. According to previous reports, the degradation pathway of LVFX may include the processes of piperazine ring oxidation, quinolone moieties transformation and decarboxylation [54,55]. Combining with the identified

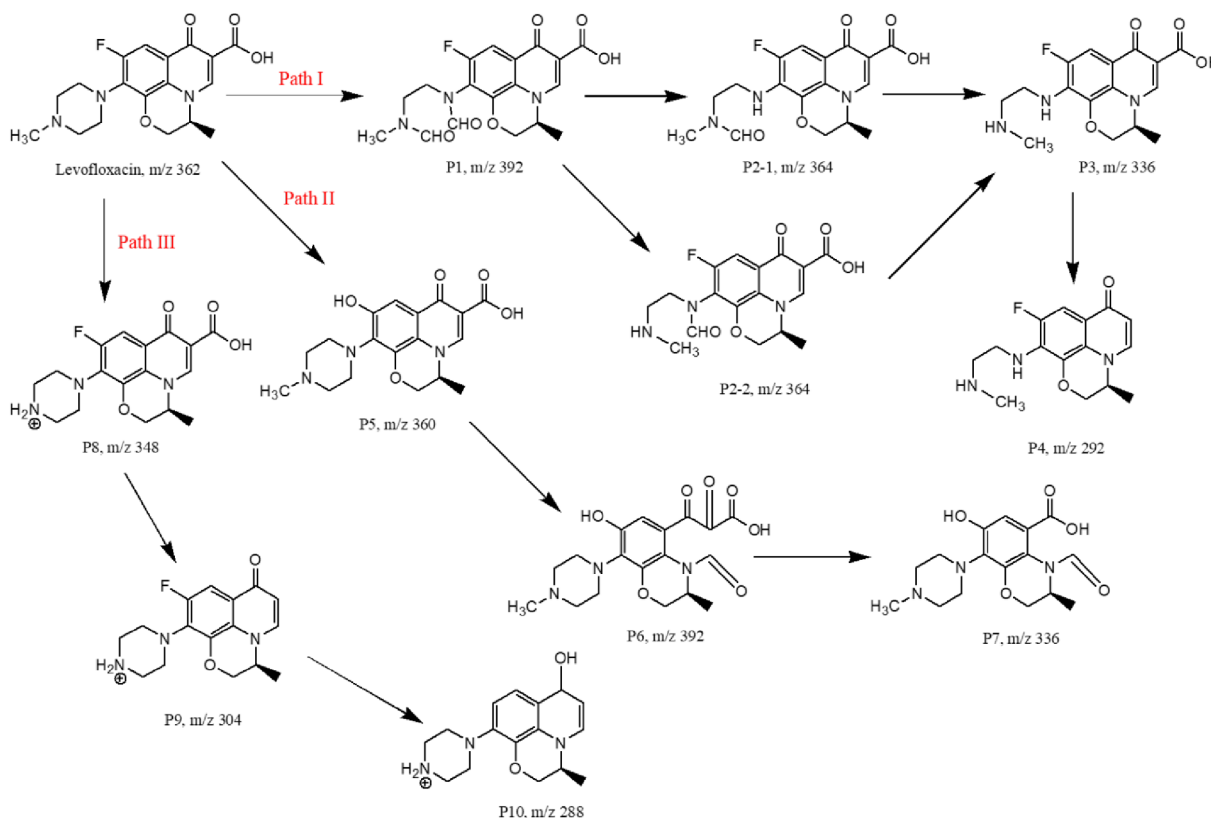


Fig. 9. Proposed pathways for LVFX degradation by $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite under visible light irradiation.

intermediate products, three possible photocatalytic degradation pathways of LVFX are put forward in our study. As shown in Fig. 9, in Path I, under the attack of reactive groups, the piperazine ring is oxidized and further broken to form P1. Some of partially de-alkylated intermediates can be further oxidized into aldehyde products P2-1, P2-2 and P3 [56]. Ultimately, P4 is generated via decarboxylation. Path II mainly refers to the destruction of quinolone moieties. Originally, the F group in LVFX is substituted with a hydroxyl group. And due to the strong oxidizing ability of the reactive groups, the quinolone ring can be easily destroyed and produce the compound P6 and P7 [2]. The third reaction mechanism is demethylation of LVFX to form P8, and it can be further decarboxylation, defluorination and hydroxylation to form P10 [57]. All these intermediates might be further broken into small molecule organic acids as well as other inorganic compounds.

Moreover, three-dimensional excitation-emission matrix fluorescence spectra (3D EEMs) technology was applied to make a deep understanding for the LVFX degradation process. LVFX, as one of fluor-quinolones (FQs), can generate fluorescence signal because of the conjugated heterocyclic structure. From Fig. 10a, the two evident fluorescence peaks at Ex/Em = 250–300/400–550 nm and Ex/Em = 300–375/400–550 nm referring to humic acid-like substances' fluorescence region can be detected, which is consistent with previous report [58]. Upon being illuminated by visible light, the fluorescence signal decreases gradually, suggesting the conjugated heterocyclic structure is broken. At the irradiation time of 90 min (Fig. 10d), there are no obvious fluorescence signals, which is attributed to the further destruction of conjugated heterocyclic structure of LVFX. The results of 3D EEMs are consistent with the above LC-MS analysis, which further demonstrates the decomposition of LVFX.

In addition, the mineralization ability of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ was also investigated to evaluate the performance of the catalyst. Fig. S4 shows the total organic carbon (TOC) remove rate of ZnFe_2O_4 , Ag_2CO_3 and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$. Generally speaking, the mineralization activity is weaker than the degradation, which may be ascribed to the formation of intermediate products during the degradation

process. As the illumination time is extended to 150 min, mineralization rate further increases, suggesting the intermediate product can be further mineralized. Among the three samples, the prepared $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite shows the highest mineralization ratio and approximately 44.75% of the TOC in the LVFX solution is removed, which indicates that the $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ nanocomposite can not only degrade LVFX but also can effectively mineralize LVFX molecules.

3.4. Catalyst stability and recyclability

In order to evaluate the recyclability and stability of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ nanocomposite, recycle experiment for degradation of LVFX under the same conditions were conducted by taking advantage of the magnetic property of the nanocomposite. As illustrated in Fig. 11a, the photocatalytic performance of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ shows no obvious deactivation after four photodegradation cycles and the degradation efficiency still maintains at about 82.56%, revealing that the photocatalyst possesses a high stability. To further explore the structure changes of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ sample during cycle experiments, XRD and TEM of used ternary catalyst were carried out. It can be seen from XRD pattern (Fig. 11b) that there is no significant change in crystal structure after four cycles. And it is observed from the TEM image (Fig. S5) that the morphology and microstructure of the used $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ sample remains the same as the fresh one. Besides, transient photocurrent response analysis of used $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ sample was applied to investigate the charges separation and recombination processes. It's clear in Fig. 11c, the photocurrent response of used $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ is still strong enough and exhibits no obvious decrease after four photodegradation cycles, which suggests the remarkable stability of the prepared photocatalyst. What's more, the magnetic properties of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ sample was depicted in Fig. 11d. Within the magnetic field range of -20000 to $20,000$ Oe, the $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite presents super-paramagnetic behavior with a saturation magnetization value of

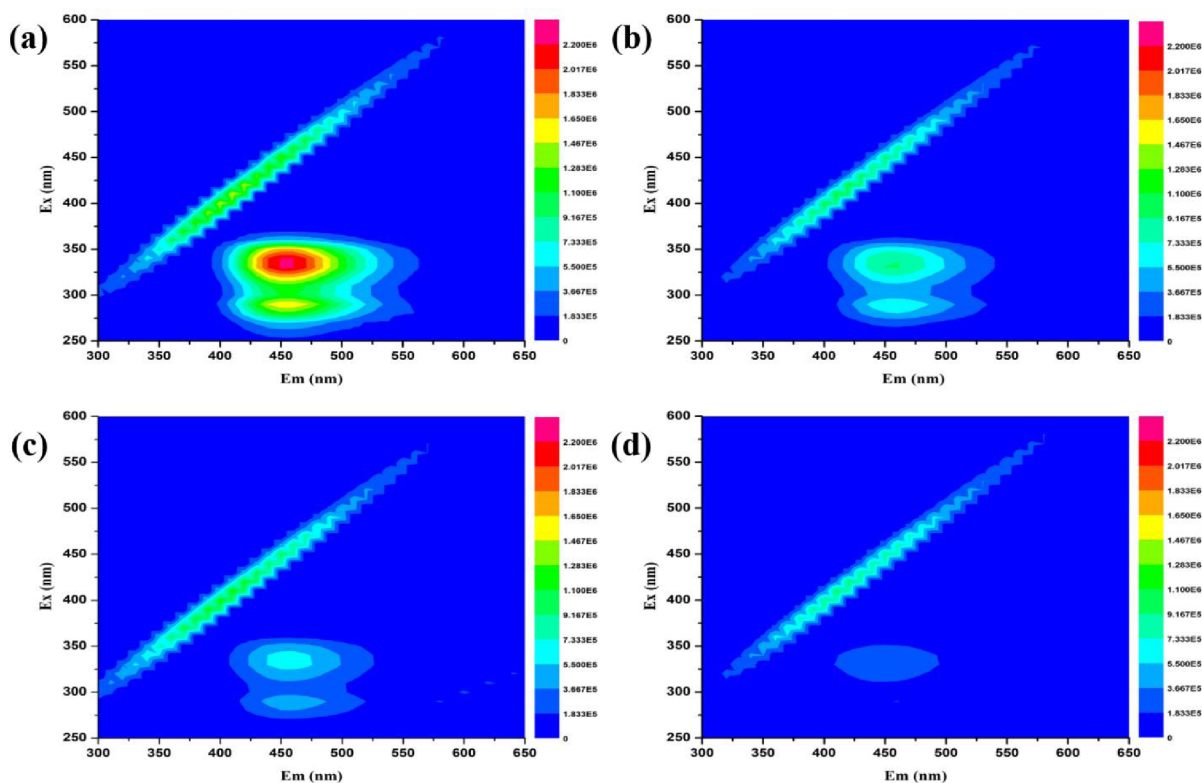


Fig. 10. 3D EEMs results of the LVFX aqueous solution with the irradiation time of: (a) 0, (b) 30, (c) 60 and (d) 90 min.

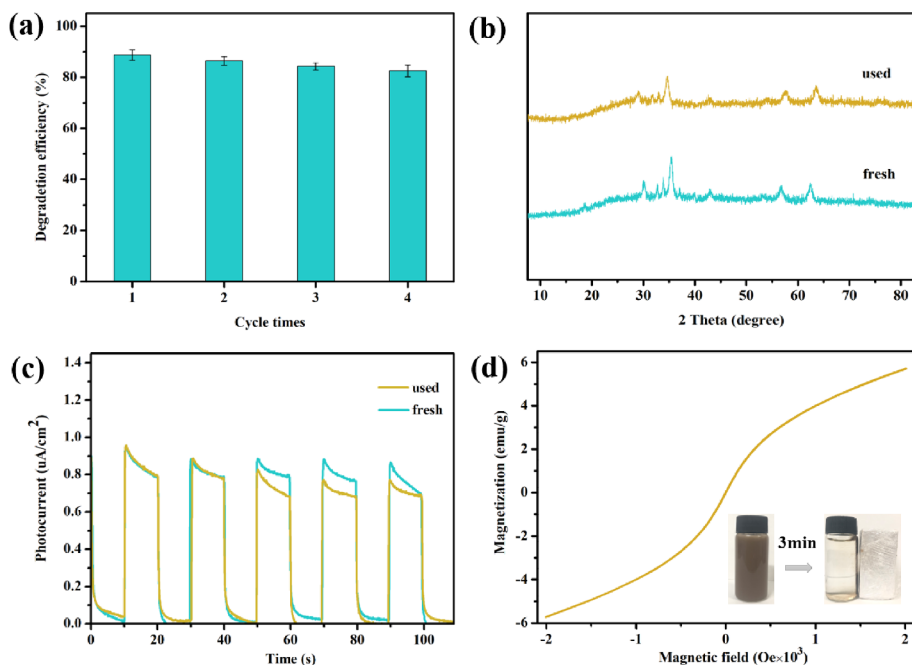


Fig. 11. (a) Circulation runs for the degradation of LVFX over $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite under visible light irradiation, (b) XRD patterns of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ before and after photocatalytic reaction, (c) transient photocurrent response of used $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ compared with the fresh one, (d) magnetic hysteresis loops of the $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite, inset: the process of separating the catalyst from the solution under an external magnetic field.

approximately 5.72 emu g^{-1} , which is strong enough to achieve efficient separation in a short time (inset in Fig. 11d). Consequently, all the results above prove that $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite holds excellent long-term stability and reusability for LVFX degradation and has a great potential in practical application.

3.5. Possible photocatalytic mechanism

To gain insight into the trend of photocatalytic performance, a series of characterizations were used to research the behaviors of charge separation and transfer. The photoelectrochemical (PC) analysis is a powerful technique to study the separation and recombination

processes of photoexcited charges. Fig. 12a exhibits the transient photocurrent responses of ZnFe_2O_4 , $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ under intermittent illumination for five on-off cycles. Notably, $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ shows the highest transient photocurrent intensity among all the samples, implying the prepared $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ nanocomposite possesses the highest separation rate of photoexcited charges. This phenomenon may be related to the synergistic effect of the formed Z-scheme transfer pathway and NCDs acting as the charge transfer medium between ZnFe_2O_4 and Ag_2CO_3 . The promoted charge transfer performance of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ was further validated by EIS measurement. It is known to us that the semicircular diameter of the EIS Nyquist diagram is related to charge-transfer

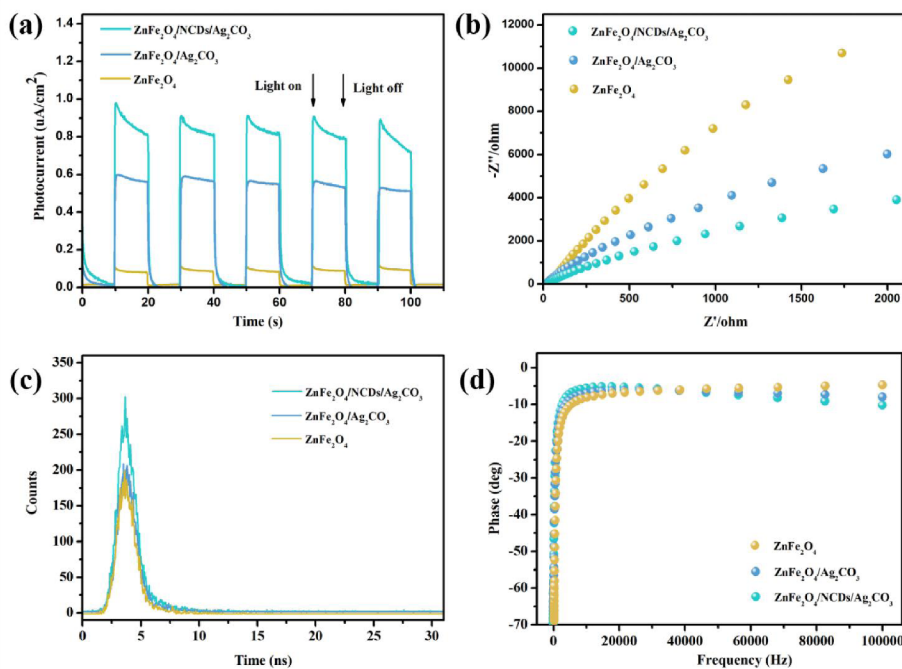


Fig. 12. (a) Transient photocurrent response, (b) EIS Nyquist plots, (c) TRPL decay spectra and (d) Bode-phase spectra of ZnFe_2O_4 , $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composites.

resistance. Generally speaking, a bigger radius implies a higher charge-transfer resistance. From Fig. 12b, it is obvious that the $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ sample has the smallest semicircle among three composites, indicating that an effective interfacial charge transfer process is achieved. This result is in good accordance with the analysis of the transient photocurrent experiment.

In addition, TRPL decay spectra of pure ZnFe_2O_4 , binary $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ were carried out to investigate the charge carrier lifetime. As shown in Fig. 12c, the average lifespan of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ is measured to be 0.8057 ns, which is longer than $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ (0.6553 ns) and ZnFe_2O_4 (0.5571 ns). That means an effective electron transport and charge separation process are achieved [7,59]. Moreover, in the Bode-phase spectra (Fig. 12d), the gradually lowered frequency peak of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ further conforms the prolonged electron lifetime, which is related to enhanced interface charge separation [24]. In brief, all above analysis demonstrate that the introduction of Ag_2CO_3 and NCDs can obviously improve the charge transfer and separation in $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ photocatalytic system, thus promote the photocatalytic performance.

It is known that a highly efficient photocatalytic system is inseparable from the generation of reactive groups. For investigating the predominate active species for LVFX degradation in photocatalytic process, the radical trapping experiments were carried out by using BQ, IPA and $\text{Na}_2\text{C}_2\text{O}_4$ as scavenger for $\cdot\text{O}_2^-$, $\cdot\text{OH}$ and h^+ , respectively. From Fig. 13a, we can learn that after putting BQ, IPA and $\text{Na}_2\text{C}_2\text{O}_4$ into photocatalytic reaction system, the photoactivity of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite is inhibited obviously and the degradation efficiencies decrease to 44.46%, 69.98% and 55.82%, respectively. This result indicates $\cdot\text{O}_2^-$, $\cdot\text{OH}$ and h^+ are all produced and participated in the degradation of LVFX.

For further proving the generation of active species in this photocatalytic system, ESR spintrap with DMPO technique under visible light irradiation was also conducted on $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ catalyst. Fig. 13c shows that there is no apparent signal for $\text{DMPO}\cdot\cdot\text{O}_2^-$ in the darkness. However, the characteristic peaks of $\text{DMPO}\cdot\cdot\text{O}_2^-$ are detected in the two composites when illuminated under visible light. Additionally, four obvious peaks of $\text{DMPO}\cdot\cdot\text{OH}$ are also measured when exposed to visible light. And the peak intensity of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composites is stronger than the binary one, indicating the promoted $\cdot\text{O}_2^-$ and $\cdot\text{OH}$ generation ability of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$.

Combining with the previous characterization and experiment, the superior $\cdot\text{O}_2^-$ and $\cdot\text{OH}$ generation ability could be attributed to the broadened optical absorption and enhanced charge separation after the introduction of NCDs. The results reveal that both $\cdot\text{O}_2^-$ and $\cdot\text{OH}$ are generated in the photocatalytic process, that is in line with analysis recorded in free radical trapping experiments. Consequently, from the radicals trap experiments and ESR analysis, we can know that the photocatalytic system is powered by the synergetic effect of $\cdot\text{OH}$, h^+ and $\cdot\text{O}_2^-$.

In addition, the Mott-Schottky (M-S) test was used to calculate the flat band potential (E_{fb}) of single ZnFe_2O_4 and Ag_2CO_3 . Generally speaking, the flat potential (E_{fb}) can be calculated through the equation [60]:

$$\frac{1}{C^2} = \frac{2}{q\epsilon\epsilon_0 N_d} \left(E - E_{\text{fb}} - \frac{k_b T}{q} \right)$$

where ϵ_0 and ϵ represent the electric permittivity of the vacuum and the dielectric constant of the used sample, and N_d , C , k_b , T , E , q denote donor density, space charge capacitance, Boltzmann's constant, Kelvin temperature, applied potential and electronic charge, respectively. As shown in Fig. 14, the E_{fb} of ZnFe_2O_4 and Ag_2CO_3 are measured to be -0.91 and 0.25 eV (vs. Ag/AgCl), corresponding to -0.71 and 0.45 eV (vs. NHE). What's more, the positive slopes of the M-S plots indicate the as-prepared ZnFe_2O_4 and Ag_2CO_3 are both n-type semiconductors. As a matter of fact, it's recorded that the E_{fb} potential of an n-type semiconductor is lower than the potential of conduction band position (E_{CB}) by 0.1 eV [61]. Hence, the E_{CB} of pristine ZnFe_2O_4 and Ag_2CO_3 are identified as -0.81 and 0.35 eV. Since the band gap energies (E_g) of ZnFe_2O_4 and Ag_2CO_3 are measured to be 1.87 and 2.57 eV corresponding to the results of DRS, their valence band (E_{VB}) potential can be estimated to be 1.06 and 2.92 eV by using the following formula as: $E_{\text{VB}} = E_{\text{CB}} + E_g$.

Based on above characterization analysis and discussion, a possible Z-scheme photocatalytic mechanism of NCDs and Ag_2CO_3 co-modified ZnFe_2O_4 ternary photocatalyst was put forward and illustrated in Scheme 1. Upon exposure to visible light, both of ZnFe_2O_4 and Ag_2CO_3 are able to be excited to produce photoinduced charges because of their narrow bandgaps. Simultaneously, NCDs can act as charge transport mediator between ZnFe_2O_4 and Ag_2CO_3 . The photogenerated electrons from the CB of Ag_2CO_3 will transfer to the surface of NCDs and then

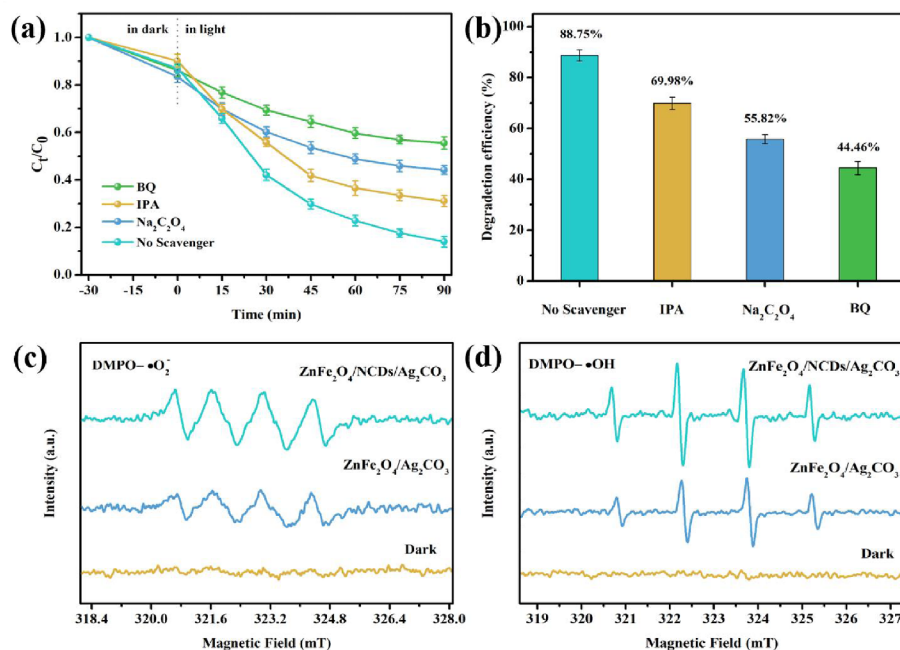


Fig. 13. (a) Photocatalytic performance of $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ composite with different trapping agents under visible-light irradiation (b) corresponding removal efficiency for LVFX with different trapping agents. DMPO spin-trapping ESR spectra over the $\text{ZnFe}_2\text{O}_4/\text{NCDs}/\text{Ag}_2\text{CO}_3$ and $\text{ZnFe}_2\text{O}_4/\text{Ag}_2\text{CO}_3$ composites: (c) $\text{DMPO}\cdot\cdot\text{O}_2^-$ response and (d) $\text{DMPO}\cdot\cdot\text{OH}$ response.

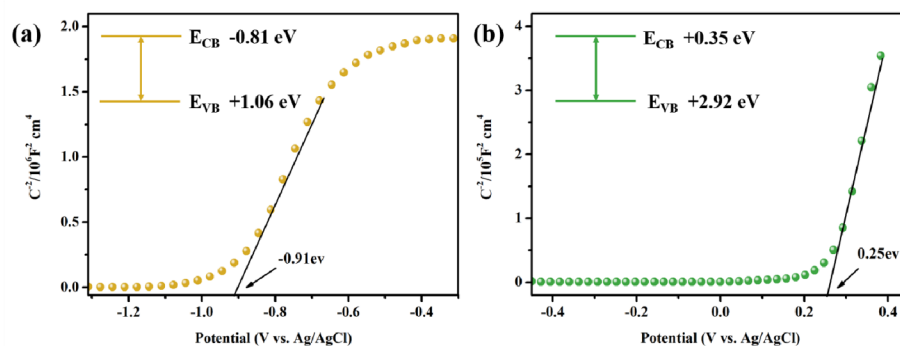
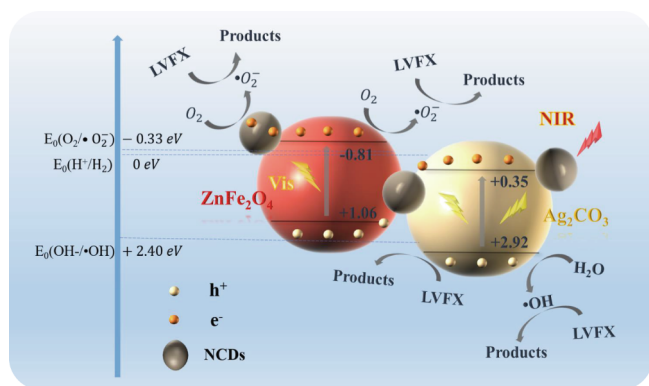


Fig. 14. Mott-Schottky curves of (a) ZnFe_2O_4 and (b) Ag_2CO_3 .



Scheme 1. Illustration of speculated degradation paths of LVFX over ZnFe_2O_4 /NCDs/ Ag_2CO_3 photocatalyst.

integrate with the holes in VB of ZnFe_2O_4 , leaving the electrons in CB of ZnFe_2O_4 and holes in VB of Ag_2CO_3 accumulated. Moreover, according to the analysis of M-S measurement, the CB potential of ZnFe_2O_4 (-0.81 eV) is more negative than the reduction potential of $\text{O}_2/\cdot\text{O}_2^-$ (-0.33 eV/vs. NHE) and the VB potential of Ag_2CO_3 (2.92 eV) is more positive than the potential of $\text{OH}^-/\cdot\text{OH}$ ($+2.40$ eV/vs. NHE), so the photoexcited holes in VB of Ag_2CO_3 can react with H_2O to generate $\cdot\text{OH}$ while electrons in the CB of ZnFe_2O_4 are able to reduce O_2 into $\cdot\text{O}_2^-$. Consequently, more active groups are generated, which are responsible for the high degradation efficiency of LVFX molecules. It is worth noting that NCDs become an important and indispensable part in the photocatalytic reaction system for the following reasons: (i) NCDs can serve as the charge transfer bridge between ZnFe_2O_4 and Ag_2CO_3 due to their remarkable electrical conductivity, which facilitates the recombination of holes from VB of ZnFe_2O_4 and electrons from CB of Ag_2CO_3 , thereby favoring effective charge separation as well as higher photocatalytic performance. (ii) As a photosensitizer, NCDs bring wider light absorption for the ternary catalytic system owing to their unique upconversion photoluminescence properties, therefore more available charge carriers are formed by the excited ZnFe_2O_4 and Ag_2CO_3 and a boosted photocatalytic activity is achieved. (iii) NCDs are able to accept the redundant electrons from ZnFe_2O_4 and Ag_2CO_3 , which is another method to advance the separation of photoinduced charges. Moreover, the electrons transferred from ZnFe_2O_4 possess a high reduction energy and can reduce O_2 to produce more $\cdot\text{O}_2^-$ active species.

4. Conclusion

In general, the magnetically separable ZnFe_2O_4 /NCDs/ Ag_2CO_3 all-solid-state Z-scheme nanocomposite was synthesized by an easy-operated deposition-precipitation route. The obtained ZnFe_2O_4 /NCDs/ Ag_2CO_3 nanocomposite exhibits an enhanced photocatalytic activity for LVFX removal, achieving 88.75% and 22.53% degradation efficiency

under visible and near-infrared (NIR) light irradiation, respectively. The promoted photocatalytic performance of the as-prepared ZnFe_2O_4 /NCDs/ Ag_2CO_3 nanocomposite is attributed to the high separation rate of photoinduced charges as well as wider optical absorption, resulting from the construction of Z-scheme heterojunction and introduction of NCDs. In addition, the results of radical trapping experiments and ESR technology confirm that $\cdot\text{O}_2^-$, $\cdot\text{OH}$ and h^+ are all produced in the photocatalytic process and $\cdot\text{O}_2^-$ plays major role for LVFX degradation among three active groups. Furthermore, the proposed ZnFe_2O_4 /NCDs/ Ag_2CO_3 hybrid composite can be recovered and shows remarkable photostability and reusability. This research provides a novel insight for the fabrication of all-solid-state Z-scheme photocatalyst based on nitrogen doped carbon quantum dots and opens new vista for designing the magnetically separable and highly efficient photocatalysts for application in wastewater remediation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2019.123192>.

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